SI252 Reinforcement Learning 2026/09/16

Homework 1

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Professor: Ziyu Shao *Due: 2026/09/30 10:59pm*

1. First, illustrate the motivation behind introducing the concept of a random variable. Then, explain why a random variable, which is in fact a mapping (i.e., a function), is nevertheless called a random variable, even though the term itself suggests that it is a variable.

   Solution:

   Motivation. The sample space Ω contains every possible outcome, and those outcomes need not be numbers: A random variable is the device that extracts such a number. Formally, a random variable is a measurable function

   $$X : \Omega \to \mathbb{R},$$

   so that for each outcome $\omega \in \Omega$ the value $X(\omega)$ is determined. Once X is introduced, probabilities, expectations, and distributions can be discussed in terms of the number X rather than in terms of the often complicated set Ω.

   Why a function is called a variable. The mapping X itself is deterministic: given ω, the number $X(\omega)$ is fixed. The randomness lies entirely in which outcome ω occurs. Before the experiment is performed, ω is unknown, so the numerical value $X(\omega)$ is unknown and changes from one run of the experiment to another. From the experimenter's point of view, X therefore behaves as a quantity that takes values at random. Thus "variable" refers to the randomly realized value, while the mathematical object that produces that value is a function of the outcome.

2. Show the proof of following results

   (a) General Bayes' Rule (four cases).

	Y discrete	Y continuous
X discrete	$P(Y = y \mid X = x) = \dfrac{P(X = x \mid Y = y)\, P(Y = y)}{P(X = x)}$	$f_Y(y \mid X = x) = \dfrac{P(X = x \mid Y = y)\, f_Y(y)}{P(X = x)}$
X continuous	$P(Y = y \mid X = x) = \dfrac{f_X(x \mid Y = y)\, P(Y = y)}{f_X(x)}$	$f_{Y\mid X}(y \mid x) = \dfrac{f_{X\mid Y}(x \mid y)\, f_Y(y)}{f_X(x)}$

Solution.

(i) X discrete, Y discrete:

$$P(Y = y \mid X = x)\, P(X = x) = P(X = x, Y = y) = P(X = x \mid Y = y)\, P(Y = y),$$

$$P(Y = y \mid X = x) = \frac{P(X = x \mid Y = y)\, P(Y = y)}{P(X = x)}.$$

(ii) X discrete, Y continuous:

$$f_Y(y \mid X = x)\, P(X = x) = f_{X,Y}(x, y) = P(X = x \mid Y = y)\, f_Y(y),$$

$$f_Y(y \mid X = x) = \frac{P(X = x \mid Y = y)\, f_Y(y)}{P(X = x)}.$$

(iii) X continuous, Y discrete:

$$f_X(x)\, P(Y = y \mid X = x) = f_{X,Y}(x, y) = f_X(x \mid Y = y)\, P(Y = y),$$

$$P(Y = y \mid X = x) = \frac{f_X(x \mid Y = y)\, P(Y = y)}{f_X(x)}.$$

(iv) X continuous, Y continuous:

$$f_{Y\mid X}(y \mid x)\, f_X(x) = f_{X,Y}(x, y) = f_{X\mid Y}(x \mid y)\, f_Y(y),$$

$$f_{Y\mid X}(y \mid x) = \frac{f_{X\mid Y}(x \mid y)\, f_Y(y)}{f_X(x)}.$$

(b) General LOTP (four cases).

	Y discrete	Y continuous
X discrete	$P(X=x)=\sum_y P(X=x \mid Y=y)\,P(Y=y)$	$P(X=x)=\int_{-\infty}^{\infty} P(X=x \mid Y=y)\,f_Y(y)\,dy$
X continuous	$f_X(x)=\sum_y f_X(x \mid Y=y)\,P(Y=y)$	$f_X(x)=\int_{-\infty}^{\infty} f_{X\mid Y}(x \mid y)\,f_Y(y)\,dy$

Solution.

(i) X discrete, Y discrete:

$$\begin{aligned} P(X=x) &= \sum_y P(X=x, Y=y) \\ &= \sum_y P(X=x \mid Y=y)\,P(Y=y). \end{aligned}$$

(ii) X discrete, Y continuous:

$$\begin{aligned} P(X=x) &= \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy \\ &= \int_{-\infty}^{\infty} P(X=x \mid Y=y)\,f_Y(y)\,dy. \end{aligned}$$

(iii) X continuous, Y discrete:

$$\begin{aligned} f_X(x) &= \sum_y f_{X,Y}(x,y) \\ &= \sum_y f_X(x \mid Y=y)\,P(Y=y). \end{aligned}$$

(iv) X continuous, Y continuous:

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy \\ &= \int_{-\infty}^{\infty} f_{X\mid Y}(x \mid y)\,f_Y(y)\,dy. \end{aligned}$$

(c) Law of Total Variance. For any r.v.s X and Y,

$$\mathrm{Var}(Y) = E\big(\mathrm{Var}(Y \mid X)\big) + \mathrm{Var}\big(E(Y \mid X)\big).$$

Solution.

$$\begin{aligned} E\big[(Y - E(Y \mid X))(E(Y \mid X) - E(Y))\big] &= E\Big[\big(E(Y \mid X) - E(Y)\big)\, E\big(Y - E(Y \mid X) \mid X\big)\Big] \\ &= 0. \end{aligned}$$

$$\begin{aligned} \mathrm{Var}(Y) &= E\big[(Y - E(Y))^2\big] \\ &= E\big[\big((Y - E(Y \mid X)) + (E(Y \mid X) - E(Y))\big)^2\big] \\ &= E\big[(Y - E(Y \mid X))^2\big] + E\big[(E(Y \mid X) - E(Y))^2\big] \\ &= E\big[E\big((Y - E(Y \mid X))^2 \mid X\big)\big] + \mathrm{Var}\big(E(Y \mid X)\big) \\ &= E\big(\mathrm{Var}(Y \mid X)\big) + \mathrm{Var}\big(E(Y \mid X)\big). \end{aligned}$$

(d) The Linear Least Square Estimate (LLSE) of Y given X, denoted by $L[Y \mid X]$, is the linear function $a + bX$ that minimizes $E[(Y - a - bX)^2]$. In fact,

$$L[Y \mid X] = E(Y) + \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)} (X - E(X)).$$

Solution. Assume $\mathrm{Var}(X) > 0$. Let $J(a, b) = E[(Y - a - bX)^2]$. Then

$$\begin{aligned} \frac{\partial J}{\partial a} &= -2E(Y - a - bX) = 0, \\ \frac{\partial J}{\partial b} &= -2E[(Y - a - bX)X] = 0, \end{aligned}$$

so

$$\begin{aligned} a + bE(X) &= E(Y), \\ E[(Y - a - bX)X] &= 0. \end{aligned}$$

Hence $a = E(Y) - bE(X)$. Substituting gives

$$\begin{aligned} E\big[(Y - E(Y) - b(X - E(X)))X\big] &= 0, \\ b &= \frac{E[(Y - E(Y))X]}{E[(X - E(X))X]}. \end{aligned}$$

$$\begin{aligned} E[(Y - E(Y))X] &= E[(Y - E(Y))(X - E(X))] = \mathrm{Cov}(X, Y), \\ E[(X - E(X))X] &= E[(X - E(X))^2] = \mathrm{Var}(X). \end{aligned}$$

Thus $b = \mathrm{Cov}(X, Y)/\mathrm{Var}(X)$, and

$$\begin{aligned} L[Y \mid X] &= a + bX \\ &= E(Y) - bE(X) + bX \\ &= E(Y) + \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)} (X - E(X)). \end{aligned}$$

(e) Orthogonality Property of MMSE. For any function $\phi(\cdot)$, one has

$$E\big[(Y - E[Y \mid X])\,\phi(X)\big] = 0.$$

Moreover, if the function $g(X)$ is such that

$$E\big[(Y - g(X))\,\phi(X)\big] = 0, \quad \forall \phi(\cdot),$$

then $g(X) = E(Y \mid X)$.

Solution.

$$\begin{aligned} E\big[(Y - E[Y \mid X])\,\phi(X)\big] &= E\Big[\phi(X)\,E\big(Y - E[Y \mid X] \mid X\big)\Big] \\ &= 0. \end{aligned}$$

If $E[(Y - g(X))\phi(X)] = 0$ for every ϕ, then

$$\begin{aligned} E\big[(E[Y \mid X] - g(X))\,\phi(X)\big] &= E\big[(Y - g(X))\phi(X)\big] - E\big[(Y - E[Y \mid X])\phi(X)\big] \\ &= 0. \end{aligned}$$

Take $\phi(X) = E[Y \mid X] - g(X)$:

$$E\big[(E[Y \mid X] - g(X))^2\big] = 0,$$

so $g(X) = E[Y \mid X]$ almost surely.

(f) Let Z be a random variable with finite variance σ^2. Then for any constant $a > 0$,

$$P(|Z - E[Z]| \geq a) \leq \frac{2\sigma^2}{\sigma^2 + a^2}.$$

Solution. Let $W = Z - E[Z]$, so $E(W) = 0$ and $\mathrm{Var}(W) = \sigma^2$. For any $\lambda > 0$,

$$\begin{aligned} P(W \geq a) &= P(W + \lambda \geq a + \lambda) \\ &\leq P((W + \lambda)^2 \geq (a + \lambda)^2) \\ &\leq \frac{E[(W + \lambda)^2]}{(a + \lambda)^2} \\ &= \frac{\sigma^2 + \lambda^2}{(a + \lambda)^2}. \end{aligned}$$

Take $\lambda = \sigma^2/a$:

$$\begin{aligned} \frac{\sigma^2 + \lambda^2}{(a + \lambda)^2} &= \frac{\sigma^2 + \sigma^4/a^2}{(a + \sigma^2/a)^2} \\ &= \frac{\sigma^2(a^2 + \sigma^2)/a^2}{(a^2 + \sigma^2)^2/a^2} \\ &= \frac{\sigma^2}{\sigma^2 + a^2}. \end{aligned}$$

The same bound holds for $-W$. The two events are disjoint, so

$$\begin{aligned} P(|W| \geq a) &= P(W \geq a) + P(-W \geq a) \\ &\leq \frac{2\sigma^2}{\sigma^2 + a^2}. \end{aligned}$$

(g) General Hoeffding Inequality. Let the random variables $X_1, X_2, \ldots, X_n$ be independent, with $a_k \le X_k \le b_k$ for each k, where a_k, b_k are constants. Let $S_n = \sum_{k=1}^n X_k$ and let $\mu = E(S_n)$. Then for any $t \ge 0$,

$$P(|S_n - \mu| \ge t) \le 2\exp\left(-\frac{2t^2}{\sum_{k=1}^n (b_k - a_k)^2}\right).$$

Solution.
Let $Y_k = X_k - E(X_k)$ and $\Sigma = \sum_{k=1}^n (b_k - a_k)^2$. Then $E(Y_k) = 0$, $S_n - \mu = \sum_{k=1}^n Y_k$, and each Y_k lies in an interval of length $b_k - a_k$. The case $t = 0$ is immediate. If $\Sigma = 0$, then $S_n = \mu$ almost surely, so the probability is 0 for $t > 0$. Now assume $t > 0$ and $\Sigma > 0$. For $\lambda > 0$, independence and Hoeffding's lemma give

$$\begin{aligned} P(S_n - \mu \ge t) &\le e^{-\lambda t}\, E\big[e^{\lambda(S_n - \mu)}\big] \\ &= e^{-\lambda t} \prod_{k=1}^n E\big[e^{\lambda Y_k}\big] \\ &\le e^{-\lambda t} \prod_{k=1}^n \exp\Big(\frac{1}{8}\lambda^2 (b_k - a_k)^2\Big) \\ &= \exp\Big(-\lambda t + \frac{\lambda^2}{8}\Sigma\Big). \end{aligned}$$

Take $\lambda = 4t/\Sigma$:

$$\begin{aligned} -\lambda t + \frac{\lambda^2}{8}\Sigma &= -\frac{4t^2}{\Sigma} + \frac{1}{8} \cdot \frac{16t^2}{\Sigma^2} \cdot \Sigma \\ &= -\frac{2t^2}{\Sigma}. \end{aligned}$$

Thus $P(S_n - \mu \ge t) \le e^{-2t^2/\Sigma}$. The same bound holds for $\mu - S_n$, so

$$P(|S_n - \mu| \ge t) \le 2\exp\Big(-\frac{2t^2}{\Sigma}\Big) = 2\exp\Big(-\frac{2t^2}{\sum_{k=1}^n (b_k - a_k)^2}\Big).$$

3. Let $X_1 \sim \text{Expo}(\lambda_1)$, $X_2 \sim \text{Expo}(\lambda_2)$ and $X_3 \sim \text{Expo}(\lambda_3)$ be independent. Find $E(X_1 \mid X_1 > 2026)$, $E(X_1 \mid X_1 < 2026)$ and $E(X_1 + X_2 + X_3 \mid X_1 > 1997,\, X_2 > 2014,\, X_3 > 2026)$.

Solution. Take $\text{Expo}(\lambda)$ to have density $\lambda e^{-\lambda x}$ for $x > 0$. For $c > 0$,

$$\begin{aligned}\int_c^\infty x\lambda e^{-\lambda x}\,dx &= \left[-xe^{-\lambda x}\right]_c^\infty + \int_c^\infty e^{-\lambda x}\,dx \\ &= ce^{-\lambda c} + \frac{1}{\lambda}e^{-\lambda c}.\end{aligned}$$

Since $P(X_1 > c) = e^{-\lambda_1 c}$,

$$\begin{aligned}E(X_1 \mid X_1 > 2026) &= \frac{2026e^{-\lambda_1 \cdot 2026} + \lambda_1^{-1}e^{-\lambda_1 \cdot 2026}}{e^{-\lambda_1 \cdot 2026}} \\ &= 2026 + \frac{1}{\lambda_1}.\end{aligned}$$

Likewise,

$$\begin{aligned}\int_0^{2026} x\lambda_1 e^{-\lambda_1 x}\,dx &= \left[-xe^{-\lambda_1 x}\right]_0^{2026} + \int_0^{2026} e^{-\lambda_1 x}\,dx \\ &= -2026e^{-\lambda_1 \cdot 2026} + \frac{1 - e^{-\lambda_1 \cdot 2026}}{\lambda_1}.\end{aligned}$$

With $P(X_1 < 2026) = 1 - e^{-\lambda_1 \cdot 2026}$,

$$\begin{aligned}E(X_1 \mid X_1 < 2026) &= \frac{1}{\lambda_1} - \frac{2026e^{-\lambda_1 \cdot 2026}}{1 - e^{-\lambda_1 \cdot 2026}} \\ &= \frac{1}{\lambda_1} - \frac{2026}{e^{\lambda_1 \cdot 2026} - 1}.\end{aligned}$$

Independence gives $E(X_i \mid X_1 > 1997,\, X_2 > 2014,\, X_3 > 2026) = E(X_i \mid X_i > c_i)$, where $c_1 = 1997$, $c_2 = 2014$, $c_3 = 2026$. Therefore

$$\begin{aligned}&E(X_1 + X_2 + X_3 \mid X_1 > 1997,\, X_2 > 2014,\, X_3 > 2026) \\ &\quad= \left(1997 + \frac{1}{\lambda_1}\right) + \left(2014 + \frac{1}{\lambda_2}\right) + \left(2026 + \frac{1}{\lambda_3}\right) \\ &\quad= 6037 + \frac{1}{\lambda_1} + \frac{1}{\lambda_2} + \frac{1}{\lambda_3}.\end{aligned}$$

4. Let Z_1, Z_2 be two i.i.d. random variables satisfying standard normal distributions, i.e., $Z_1, Z_2 \sim \mathcal{N}(0,1)$. Define

$$X = \sigma_X Z_1 + \mu_X,$$
$$Y = \sigma_Y\left(\rho Z_1 + \sqrt{1-\rho^2}\, Z_2\right) + \mu_Y,$$

where $\sigma_X > 0$, $\sigma_Y > 0$, $-1 < \rho < 1$.

(a) Show that X and Y are bivariate normal.
Solution. For any $a, b \in \mathbb{R}$,

$$aX + bY = a\mu_X + b\mu_Y + (a\sigma_X + b\sigma_Y\rho)Z_1 + b\sigma_Y\sqrt{1-\rho^2}\, Z_2.$$

The right-hand side is an affine function of the independent standard normals Z_1, Z_2, hence normal. Every linear combination of X and Y is normal, so (X, Y) is bivariate normal.

(b) Find the correlation coefficient between X and Y, i.e., $\mathrm{Corr}(X, Y)$.
Solution. Since Z_1, Z_2 are independent and $\mathrm{Var}(Z_i) = 1$,

$$\begin{aligned}
\mathrm{Var}(X) &= \sigma_X^2, \\
\mathrm{Var}(Y) &= \sigma_Y^2(\rho^2 + (1-\rho^2)) = \sigma_Y^2, \\
\mathrm{Cov}(X, Y) &= \mathrm{Cov}(\sigma_X Z_1,\, \sigma_Y\rho Z_1 + \sigma_Y\sqrt{1-\rho^2}\, Z_2) \\
&= \sigma_X\sigma_Y\rho.
\end{aligned}$$

$$\mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}} = \rho.$$

(c) Find the joint PDF of X and Y.
Solution. The joint density of (Z_1, Z_2) is $(2\pi)^{-1}\exp\bigl(-(z_1^2 + z_2^2)/2\bigr)$. Invert the transformation:

$$Z_1 = \frac{X - \mu_X}{\sigma_X},$$
$$Z_2 = \frac{1}{\sqrt{1-\rho^2}}\left(\frac{Y - \mu_Y}{\sigma_Y} - \rho\frac{X - \mu_X}{\sigma_X}\right).$$

The Jacobian matrix of $(Z_1, Z_2) \mapsto (X, Y)$ is

$$\begin{pmatrix} \sigma_X & 0 \\ \sigma_Y\rho & \sigma_Y\sqrt{1-\rho^2} \end{pmatrix},$$

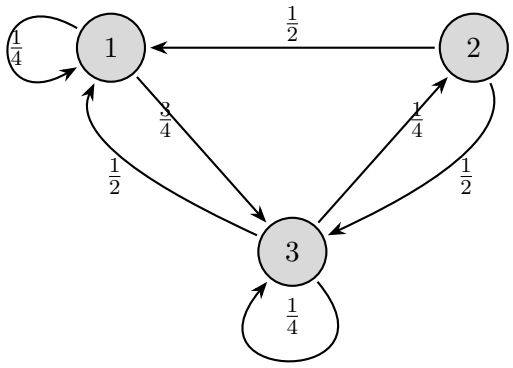
with determinant $\sigma_X\sigma_Y\sqrt{1-\rho^2}$. Writing $U = (X - \mu_X)/\sigma_X$ and $V = (Y - \mu_Y)/\sigma_Y$,

$$\begin{aligned}
Z_1^2 + Z_2^2 &= U^2 + \frac{(V - \rho U)^2}{1-\rho^2} \\
&= \frac{U^2(1-\rho^2) + (V - \rho U)^2}{1-\rho^2} \\
&= \frac{U^2 + V^2 - 2\rho UV}{1-\rho^2} \\
&= \frac{1}{1-\rho^2}\left(\frac{(X - \mu_X)^2}{\sigma_X^2} + \frac{(Y - \mu_Y)^2}{\sigma_Y^2} - 2\rho\frac{(X - \mu_X)(Y - \mu_Y)}{\sigma_X\sigma_Y}\right).
\end{aligned}$$

Therefore, for all $x, y \in \mathbb{R}$,

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \times \exp\left(-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_X)^2}{\sigma_X^2} + \frac{(y-\mu_Y)^2}{\sigma_Y^2} - 2\rho\frac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y}\right]\right).$$

5. Given a time-homogeneous Markov chain with state-transition diagram shown as follows:



Ordering the states as $1, 2, 3$, the transition matrix is

$$P = \begin{pmatrix} 1/4 & 0 & 3/4 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/4 & 1/4 \end{pmatrix}.$$

(a) Find $P(X_5 = 3 \mid X_4 = 2)$ and $P(X_4 = 1 \mid X_3 = 2)$.
Solution. By time-homogeneity and the Markov property,

$$P(X_5 = 3 \mid X_4 = 2) = P_{23} = \frac{1}{2},$$
$$P(X_4 = 1 \mid X_3 = 2) = P_{21} = \frac{1}{2}.$$

(b) Find the stationary distribution of the chain.
Solution. Solve $\pi P = \pi$ and $\pi_1 + \pi_2 + \pi_3 = 1$. The second coordinate gives

$$\pi_2 = \frac{1}{4}\pi_3.$$

The first coordinate gives

$$\pi_1 = \frac{1}{4}\pi_1 + \frac{1}{2}\pi_2 + \frac{1}{2}\pi_3,$$
$$\frac{3}{2}\pi_1 = \pi_2 + \pi_3 = 5\pi_2,$$
$$\pi_1 = \frac{10}{3}\pi_2, \qquad \pi_3 = 4\pi_2.$$

Normalization yields

$$\frac{10}{3}\pi_2 + \pi_2 + 4\pi_2 = \frac{25}{3}\pi_2 = 1,$$
$$\pi_2 = \frac{3}{25}, \qquad \pi_1 = \frac{2}{5}, \qquad \pi_3 = \frac{12}{25}.$$

Thus $\pi = \left(\frac{2}{5}, \frac{3}{25}, \frac{12}{25}\right)$.

(c) Find the period of state 2 and the period of the chain.

Solution. The paths $2 \to 3 \to 2$ and $2 \to 3 \to 3 \to 2$ have positive probability, so the return times of state 2 include 2 and 3. Their gcd is 1, hence the period of state 2 is 1.

The chain is irreducible: $1 \to 3 \to 2 \to 1$. An irreducible chain has a common period, so the period of the chain is 1.

(d) If $P(X_0 = 3) = \dfrac{1}{5}$, find $P(X_0 = 3,\, X_1 = 2,\, X_2 = 1)$.

Solution.

$$\begin{aligned} P(X_0 = 3,\, X_1 = 2,\, X_2 = 1) &= P(X_0 = 3)\, P_{32}\, P_{21} \\ &= \frac{1}{5} \cdot \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{40}. \end{aligned}$$

(e) Find $E(X_5 \mid X_3 = 1)$ and $\mathrm{Var}(X_5 \mid X_3 = 1)$.

Solution. The conditional distribution of X_5 given $X_3 = 1$ is the distribution after two steps from state 1. The first row of P^2 is

$$\begin{aligned} (P^2)_{11} &= \frac{1}{4} \cdot \frac{1}{4} + \frac{3}{4} \cdot \frac{1}{2} = \frac{7}{16}, \\ (P^2)_{12} &= \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{16}, \\ (P^2)_{13} &= \frac{1}{4} \cdot \frac{3}{4} + \frac{3}{4} \cdot \frac{1}{4} = \frac{6}{16}. \end{aligned}$$

Therefore

$$\begin{aligned} E(X_5 \mid X_3 = 1) &= 1 \cdot \frac{7}{16} + 2 \cdot \frac{3}{16} + 3 \cdot \frac{6}{16} = \frac{31}{16}, \\ E(X_5^2 \mid X_3 = 1) &= 1 \cdot \frac{7}{16} + 4 \cdot \frac{3}{16} + 9 \cdot \frac{6}{16} = \frac{73}{16}, \\ \mathrm{Var}(X_5 \mid X_3 = 1) &= \frac{73}{16} - \Big(\frac{31}{16}\Big)^2 = \frac{207}{256}. \end{aligned}$$

6. Given a coin with the probability p of landing heads. p is unknown and we need to estimate its value through data. In our data collection model, we have n independent tosses, result of each toss is either Head or Tail. Let X denote the number of heads in the total n tosses. Now we conduct experiments to collect data and find $X = k$. Then we need to find $\hat{p}$, the estimation of p.

   (a) Assume p is an unknown constant. Find $\hat{p}$ through the MLE (Maximum Likelihood Estimation) rule.

   Solution. $X \mid p \sim \text{Binomial}(n, p)$, so the likelihood is $L(p) = \binom{n}{k} p^k (1-p)^{n-k}$. For $0 < p < 1$,

$$\frac{d}{dp} \log L(p) = \frac{k}{p} - \frac{n-k}{1-p} = 0$$

   gives $k(1-p) = (n-k)p$, hence $\hat{p} = k/n$. The same value is the maximizer at the endpoints $k = 0$ and $k = n$.

   (b) Assume p is a random variable with a prior distribution $p \sim \text{Beta}(1, 1)$. Find $\hat{p}$ through the MAP (Maximum a Posterior Probability) rule.

   Solution. The Beta$(1, 1)$ density is constant on $(0, 1)$, so the posterior satisfies

$$f(p \mid X = k) \propto p^k (1-p)^{n-k}, \qquad 0 < p < 1,$$

   which is the Beta$(k + 1,\, n - k + 1)$ density. Its logarithm has the same derivative as $\log L(p)$, so the maximizer is again

$$\hat{p} = \frac{k}{n}.$$

   (c) Assume p is a random variable with a prior distribution $p \sim \text{Unif}(0, 1)$. Find $\hat{p}$ through the MMSE (Minimal Mean Squared Error) rule.

   Solution. Unif$(0, 1)$ = Beta$(1, 1)$, so the posterior is again Beta$(k + 1,\, n - k + 1)$. The MMSE estimator is the posterior mean:

$$\hat{p} = E(p \mid X = k) = \frac{k+1}{(k+1) + (n-k+1)} = \frac{k+1}{n+2}.$$

   (d) Assume p is a random variable with a prior distribution $p \sim \text{Unif}(0, 1)$. Find $\hat{p}$ through the LLSE (Linear Least Square Estimate) rule where

$$L[Y \mid X] = E(Y) + \frac{\text{Cov}(X, Y)}{\text{Var}(X)} (X - E(X)).$$

   Solution. Here $Y = p$. Since $p \sim \text{Unif}(0, 1)$ and $X \mid p \sim \text{Binomial}(n, p)$,

$$\begin{aligned}
E(p) = \frac{1}{2}, \qquad E(p^2) = \frac{1}{3}, \qquad \text{Var}(p) &= \frac{1}{12}, \\
E(X) = E(np) &= \frac{n}{2}, \\
\text{Var}(X) &= E\big(np(1-p)\big) + \text{Var}(np) \\
&= n\Big(\frac{1}{2} - \frac{1}{3}\Big) + n^2 \cdot \frac{1}{12} = \frac{n(n+2)}{12}, \\
\text{Cov}(X, p) &= E(np^2) - E(X)E(p) = \frac{n}{3} - \frac{n}{4} = \frac{n}{12}.
\end{aligned}$$

Therefore

$$\begin{aligned} L[p \mid X] &= \frac{1}{2} + \frac{n/12}{n(n+2)/12}\Big(X - \frac{n}{2}\Big) \\ &= \frac{1}{2} + \frac{X - n/2}{n+2} = \frac{X+1}{n+2}. \end{aligned}$$

Observing $X = k$ gives $\hat{p} = (k+1)/(n+2)$.

7. Let X, Y be jointly Gaussian random variables. Show the following equality with two different methods:

$$E[Y \mid X] = L[Y \mid X] = E(Y) + \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)}\,(X - E(X)).$$

Solution. The identity $L[Y \mid X] = E(Y) + \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)}(X - E(X))$ is Problem 2(d); assume $\mathrm{Var}(X) > 0$. It remains to show that the conditional expectation equals this linear function. Write $\mu_X = E(X)$, $\mu_Y = E(Y)$, $\sigma_X^2 = \mathrm{Var}(X)$, $\sigma_Y^2 = \mathrm{Var}(Y)$ and $\rho = \mathrm{Corr}(X,Y)$.

Method 1. If $|\rho| < 1$, Problem 4 supplies independent $Z_1, Z_2 \sim \mathcal{N}(0,1)$ such that

$$\begin{aligned} X &= \sigma_X Z_1 + \mu_X, \\ Y &= \sigma_Y\big(\rho Z_1 + \sqrt{1-\rho^2}\, Z_2\big) + \mu_Y. \end{aligned}$$

Then $Z_1 = (X - \mu_X)/\sigma_X$ is a function of X, while Z_2 is independent of X. Hence

$$\begin{aligned} E[Y \mid X] &= \sigma_Y \rho \cdot \frac{X - \mu_X}{\sigma_X} + \mu_Y \\ &= \mu_Y + \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)}(X - \mu_X) \\ &= L[Y \mid X]. \end{aligned}$$

If $|\rho| = 1$, then Y is almost surely an affine function of X, so both sides equal that function.

Method 2. Let $\widehat{Y} = L[Y \mid X]$ and $e = Y - \widehat{Y}$. The normal equations for the LLSE give $\mathrm{Cov}(e, X) = 0$, and $E(e) = 0$. The pair (e, X) is jointly Gaussian, so uncorrelated implies independent. Therefore

$$E[Y \mid X] = E[\widehat{Y} \mid X] + E[e \mid X] = \widehat{Y} + E(e) = L[Y \mid X].$$

8. Find the expectation of the returned value of the following program code:

```
f l o a t sum=0.0;
while (sum<=1.0) {
    sum+= Unif ( 0 , 1 ) ;
    }
return sum ;
```

Solution. Let $\varphi(s)$ be the expected returned value when the current sum is $s \le 1$. The program adds an independent $\mathrm{Unif}(0,1)$ random variable until the sum exceeds 1, and the answer is $\varphi(0)$. For $s \in [0,1]$,

$$\begin{aligned}\varphi(s) &= \int_0^{1-s} \varphi(s+u)\,du + \int_{1-s}^{1} (s+u)\,du \\ &= \int_s^1 \varphi(t)\,dt + s + \frac{s^2}{2}.\end{aligned}$$

Differentiating gives $\varphi'(s) = -\varphi(s) + 1 + s$, so

$$\begin{aligned}\frac{d}{ds}(\varphi(s)e^s) &= (1+s)e^s, \\ \varphi(s)e^s &= se^s + C, \\ \varphi(s) &= s + Ce^{-s}.\end{aligned}$$

If the current sum equals 1, one more uniform draw is added and the result is $1 + U$, so $\varphi(1) = 3/2$. Thus $1 + C/e = 3/2$, $C = e/2$, and

$$\varphi(0) = \frac{e}{2}.$$

9. (Challenging) Prove Hoeffding Lemma. Let the random variable X satisfy $E(X) = 0$ and $a \le X \le b$, where a and b are constants. Then for any $\lambda > 0$,

$$E\big(e^{\lambda X}\big) \le e^{\frac{1}{8}\lambda^2 (b-a)^2}.$$

Solution. If $a = b$, then $X = 0$ and both sides equal 1. Now assume $a < b$. Since $E(X) = 0$ and $a \le X \le b$, one has $a \le 0 \le b$. Convexity of $x \mapsto e^{\lambda x}$ gives, for every $x \in [a, b]$,

$$e^{\lambda x} \le \frac{b-x}{b-a}\, e^{\lambda a} + \frac{x-a}{b-a}\, e^{\lambda b}.$$

Taking expectations and using $E(X) = 0$,

$$E\big(e^{\lambda X}\big) \le \frac{b}{b-a}\, e^{\lambda a} + \frac{-a}{b-a}\, e^{\lambda b}.$$

Set $p = -a/(b-a) \in [0, 1]$, so $1 - p = b/(b-a)$. Then

$$E\big(e^{\lambda X}\big) \le (1-p)e^{\lambda a} + pe^{\lambda b} = \exp\big(\lambda a + \psi(u)\big),$$

where $u = \lambda(b-a)$ and $\psi(u) = \log(1 - p + pe^u)$. Now

$$\begin{aligned} \psi(0) &= 0, \\ \psi'(u) &= \frac{pe^u}{1-p+pe^u}, \qquad \psi'(0) = p, \\ \psi''(u) &= \psi'(u)\big(1 - \psi'(u)\big) \le \frac{1}{4}. \end{aligned}$$

Taylor expansion with the Lagrange remainder yields

$$\begin{aligned} \psi(u) &= \psi(0) + \psi'(0)\, u + \frac{1}{2}\psi''(\xi)u^2 \\ &\le pu + \frac{1}{8}u^2. \end{aligned}$$

Hence

$$\begin{aligned} \lambda a + \psi(u) &\le \lambda a + p\lambda(b-a) + \frac{1}{8}\lambda^2(b-a)^2 \\ &= \frac{1}{8}\lambda^2(b-a)^2, \end{aligned}$$

because $p(b-a) = -a$. Therefore

$$E\big(e^{\lambda X}\big) \le \exp\Big(\frac{1}{8}\lambda^2(b-a)^2\Big).$$

10. (Challenging) A resource-allocation system serves two users. Let x_1 and x_2 denote their service rates, measured in normalized units. Serving user 2 requires twice as much resource per unit of service as serving user 1. The feasible allocations are therefore

$$F = \{(x_1, x_2) \in \mathbb{R}^2_{\geq 0} : x_1 + 2x_2 \leq 12\}.$$

A system designer evaluates allocations using an aggregate utility function

$$W(x_1, x_2) = w_1 u_1(x_1) + w_2 u_2(x_2), \qquad w_1, w_2 > 0.$$

Whenever logarithmic utility is used, restrict attention to strictly positive service rates. Throughout the problem, log denotes the natural logarithm.

Central question. How does the choice of utility determine what counts as a good allocation, and how can this objective be expressed as a reinforcement-learning reward?

(a) What does a utility function represent? Explain what it means for a function u to represent a preference relation $\succeq$ over deterministic outcomes. Express your answer mathematically. Suppose that both $u(x) = x$ and $v(x) = \log x$ represent the preference "more service is better." Do they necessarily lead to the same allocation when the designer maximizes the sum of the users' utilities? Explain why the ordering of an individual user's outcomes does not, by itself, determine how outcomes should be compared across users.

Solution. A utility u represents $\succeq$ when, for all outcomes x, y,

$$x \succeq y \iff u(x) \geq u(y).$$

On $(0, \infty)$, both $u(x) = x$ and $v(x) = \log x$ are strictly increasing, so

$$x \geq y \iff u(x) \geq u(y) \iff v(x) \geq v(y).$$

They represent the same individual preference. They do not give the same social allocation. Under $x_1 + 2x_2 \leq 12$,

$$\max(x_1 + x_2) = (12, 0), \qquad \max(\log x_1 + \log x_2) = (6, 3)$$

(the second optimum is computed in (c)). An individual's ordering fixes utility only up to a strictly increasing transformation. Summing utilities compares gains across users, and that cardinal comparison is not determined by the separate orderings.

(b) What does concavity mean? Suppose that both users have the same twice-differentiable utility function u, with

$$u'(x) > 0, \qquad u''(x) < 0.$$

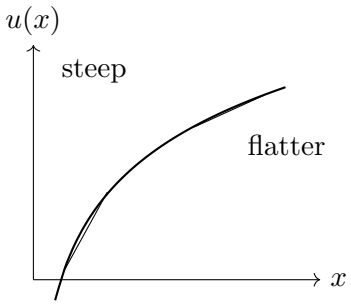
Explain the meaning of each inequality. Sketch a representative utility curve and indicate how its slope changes as service increases.

Let $0 < a < b$. Prove that transferring a small amount of service from the better-served user to the worse-served user increases aggregate utility:

$$u(a + \delta) + u(b - \delta) > u(a) + u(b), \qquad 0 < \delta < \frac{b - a}{2}.$$

Does this result imply that maximizing a sum of strictly concave utilities always gives equal service rates? Discuss the role of feasibility, using the resource constraint in this problem.

Solution. $u'(x) > 0$ means more service is strictly better. $u''(x) < 0$ means marginal utility is strictly decreasing: the curve is strictly concave and becomes flatter as service increases.



Let $g(\delta) = u(a+\delta) + u(b-\delta)$. Then $g'(\delta) = u'(a+\delta) - u'(b-\delta)$. For $0 < \delta < (b-a)/2$, one has $a + \delta < b - \delta$. Since u' is strictly decreasing, $g'(\delta) > 0$, so $g(\delta) > g(0)$.

This does not force equal service rates. The transfer above moves one unit of service from one user to the other. Under $x_1 + 2x_2 \leq 12$, that transfer need not be feasible: user 2 consumes twice as much resource per unit of service. For instance $(6, 3)$ is feasible, but $(5, 4)$ uses resource $5 + 8 = 13 > 12$. The feasible equal-rate point is $(4, 4)$, and

$$\log 4 + \log 4 = \log 16 < \log 18 = \log 6 + \log 3.$$

Strict concavity favors equalizing marginal utilities per unit of scarce resource, not the service rates themselves.

(c) Which fairness criterion corresponds to logarithmic utility? Find the optimal allocation under each of the following objectives:

$$\begin{aligned} \text{Total service:} \quad & \max_{(x_1,x_2)\in F} (x_1 + x_2), \\ \text{Logarithmic utility:} \quad & \max_{\substack{(x_1,x_2)\in F \\ x_1,x_2>0}} (\log x_1 + \log x_2), \\ \text{Minimum service:} \quad & \max_{(x_1,x_2)\in F} \min\{x_1, x_2\}. \end{aligned}$$

Compare the three allocations. Does logarithmic utility maximize total service, equalize service, or implement a different principle?

More generally, suppose that $C \subseteq \mathbb{R}^n_{>0}$ is convex and that $x^\star$ maximizes

$$W(x) = \sum_{i=1}^{n} \log x_i$$

over C. Prove that, for every $y \in C$,

$$\sum_{i=1}^{n} \frac{y_i - x_i^\star}{x_i^\star} \leq 0.$$

Interpret this inequality in words and name the corresponding fairness criterion.

Finally, find the optimal allocation when the designer uses

$$W(x_1, x_2) = 2\log x_1 + \log x_2.$$

What does the coefficient 2 represent?

Solution. Logarithmic utility implements proportional fairness, not maximum total service and not equal service.

Total service. For $(x_1, x_2) \in F$,

$$x_1 + x_2 = (x_1 + 2x_2) - x_2 \leq 12 - x_2 \leq 12,$$

with equality only at $(12, 0)$.

Logarithmic utility. The objective tends to $-\infty$ if either coordinate tends to 0, and it increases if the resource constraint has slack. Thus an optimum lies on $x_1 + 2x_2 = 12$ with $x_i > 0$. Substituting $x_1 = 12 - 2x_2$,

$$\frac{d}{dx_2}\big(\log(12 - 2x_2) + \log x_2\big) = -\frac{2}{12 - 2x_2} + \frac{1}{x_2} = 0$$

gives $x_2 = 3$ and $x_1 = 6$.

Minimum service. The largest t with $x_1 \geq t$, $x_2 \geq t$ and $x_1 + 2x_2 \leq 12$ is attained at $x_1 = x_2 = t$ and $t + 2t = 12$, so $t = 4$ and the allocation is $(4, 4)$.

Thus the three allocations are $(12, 0)$, $(6, 3)$ and $(4, 4)$. Logarithmic utility does not maximize total service ($6 + 3 = 9 < 12$) and does not equalize service. It equalizes weighted marginal utilities, $1/x_1 = (1/x_2)/2$.

Proportional fairness. Let $\varphi(t) = \sum_{i=1}^{n} \log\big((1 - t)x_i^\star + ty_i\big)$ for $t \in [0, 1]$. Convexity of C gives $\varphi(t) \leq \varphi(0)$, so $\varphi'(0) \leq 0$. Hence

$$\varphi'(0) = \sum_{i=1}^{n} \frac{y_i - x_i^\star}{x_i^\star} \leq 0.$$

No feasible move from $x^\star$ raises the sum of relative changes $(y_i - x_i^\star)/x_i^\star$. This is proportional fairness.

Weighted logarithmic utility. On $x_1 + 2x_2 = 12$,

$$\frac{d}{dx_2}\big(2\log(12 - 2x_2) + \log x_2\big) = -\frac{4}{12 - 2x_2} + \frac{1}{x_2} = 0$$

gives $x_1 = 4x_2$, so $(x_1, x_2) = (8, 2)$. The coefficient 2 is user 1's welfare weight: the objective is $\log(x_1^2 x_2)$, the Nash product in which user 1 has twice user 2's weight.

(d) Concavity under uncertainty: fairness or risk aversion? Separately, consider a single user choosing between a certain payoff and a lottery:

$$\text{Option A:} \quad X = 5 \text{ with certainty,}$$

$$\text{Option B:} \quad X = \begin{cases} 1, & \text{with probability } \dfrac{1}{2}, \\ 9, & \text{with probability } \dfrac{1}{2}. \end{cases}$$

Compare the options under the objectives $E[X]$ and $E[\log X]$. Find the certainty equivalent of Option B under logarithmic utility: the amount c satisfying

$$\log c = E[\log X].$$

Use Jensen's inequality to explain the general relationship between concavity and risk aversion. Why is this interpretation different from the fairness interpretation in part (b)?

Solution.

$$E[X] = 5 \quad \text{for both options,}$$

$$E[\log X] = \begin{cases} \log 5, & \text{Option A,} \\ \frac{1}{2}\log 1 + \frac{1}{2}\log 9 = \log 3, & \text{Option B.} \end{cases}$$

Expected payoff ties the two options. Expected log utility strictly prefers A, since $\log 3 < \log 5$. The certainty equivalent of B solves $\log c = \log 3$, so $c = 3 < 5 = E[X]$.

If u is concave, Jensen's inequality says $E[u(X)] \le u(E[X])$. An increasing concave agent therefore prefers the sure amount $E[X]$ to the lottery, and the certainty equivalent is at most $E[X]$. Concavity here is risk aversion: variability across states of the world is penalized for one user. In (b), concavity instead compares two users' deterministic service levels. The same curvature has two different meanings.

(e) The von Neumann–Morgenstern expected utility theorem. State the main content of the von Neumann–Morgenstern Expected Utility Theorem for lotteries over a finite outcome set. Your statement should identify the four standard preference axioms, the expected-utility representation, and the transformations under which the representation is unique.

Does the theorem require utility to be concave or logarithmic? Does it determine a fair allocation across users?

Explain why applying an arbitrary strictly increasing transformation to utility can preserve preferences over deterministic outcomes while changing preferences over lotteries. Use part (d) as an illustration.

Solution. For a finite outcome set, let $\succeq$ be a preference on lotteries. If it is complete, transitive, continuous, and independent, then there is a utility u on outcomes such that

$$p \succeq q \iff \sum_x p(x)\,u(x) \ge \sum_x q(x)\,u(x).$$

The four axioms are: completeness; transitivity; continuity (if $p \succ q \succ r$, some mixtures of p and r lie strictly above and below q); independence (if $p \succeq q$, then mixing both with any common lottery preserves the ranking). The representation is unique up to positive affine transformations $v = \alpha u + \beta$ with $\alpha > 0$.

The theorem does not require concavity or the logarithm; any u is allowed, and concavity is the extra assumption of risk aversion. It ranks lotteries for one decision maker. It does not compare users or select a fair allocation.

A strictly increasing φ preserves rankings of deterministic outcomes, because $\varphi(u(x)) \ge \varphi(u(y))$ if and only if $u(x) \ge u(y)$. It need not preserve rankings of lotteries, because $E[\varphi(u(X))]$ is not $\varphi(E[u(X)])$ unless φ is affine. In (d), $u(x) = x$ and $v(x) = \log x$ order sure payoffs identically, and both options have mean 5, so u is indifferent. Expected log utility prefers the sure payoff 5 to the lottery.

(f) From utility to reinforcement-learning reward. Now suppose the system operates for T time steps. A policy π generates positive allocations

$$X(t) = \big(X_1(t), X_2(t)\big) \in F, \qquad t = 0, \dots, T-1.$$

An undiscounted episodic RL agent maximizes

$$J(\pi) = E_\pi\left[\sum_{t=0}^{T-1} R_{t+1}\right].$$

State the reward hypothesis in your own words. Explain its connection to the expected utility theorem. In particular, does an expected-utility representation automatically provide a stationary Markov reward function on a prescribed state space?

Consider two possible fairness objectives:

$$J_{\text{instant}}(\pi) = E_\pi\left[\sum_{t=0}^{T-1}\left(\log X_1(t) + \log X_2(t)\right)\right],$$

and

$$J_{\text{average}}(\pi) = E_\pi[\log X_1 + \log X_2], \qquad X_i = \frac{1}{T}\sum_{t=0}^{T-1} X_i(t).$$

For $T = 2$, compare these objectives for the following deterministic schedules:

$$\pi_A : (6,3),\ (6,3),$$
$$\pi_B : (10,1),\ (2,5).$$

Are the objectives equivalent?

Construct an episodic reward whose expected cumulative sum equals $J_{\text{average}}(\pi)$. Explain what information may need to be added to the state.

Solution. The reward hypothesis says that the goals we want an agent to pursue can be expressed as maximization of the expected cumulative value of a scalar reward. Under the expected-utility theorem, a rational preference over random trajectories is an expected utility of trajectories. If that utility is a sum of per-step rewards, the criterion is exactly $J(\pi)$. An expected-utility representation does not by itself supply a stationary Markov reward on a prescribed state space: the utility may depend on the whole trajectory, and the given state may omit the statistics that make the reward Markov.

For $T = 2$,

$$J_{\text{instant}}(\pi_A) = 2(\log 6 + \log 3) = 2\log 18,$$
$$J_{\text{instant}}(\pi_B) = \log 10 + \log 1 + \log 2 + \log 5 = \log 100.$$

Since $2\log 18 = \log 324 > \log 100$, instantaneous utility strictly prefers π_A. Both schedules have averages $\overline{X}_1 = 6$ and $\overline{X}_2 = 3$, so

$$J_{\text{average}}(\pi_A) = J_{\text{average}}(\pi_B) = \log 6 + \log 3.$$

The two objectives are not equivalent.

Define $R_{t+1} = 0$ for $t = 0, \ldots, T-2$ and

$$R_T = \log\left(\frac{1}{T}\sum_{t=0}^{T-1} X_1(t)\right) + \log\left(\frac{1}{T}\sum_{t=0}^{T-1} X_2(t)\right).$$

Then $\sum_{t=0}^{T-1} R_{t+1} = \log \overline{X}_1 + \log \overline{X}_2$, so the expected cumulative reward equals $J_{\text{average}}(\pi)$. Evaluating R_T requires the time-average of each user's service. The state must therefore carry the elapsed time and the cumulative service of both users, not only the current allocation.

(g) Illustrate the connection between Von Neumann–Morgenstern Expected Utility Theorem and reward hypothesis in reinforcement learning.

Solution. The expected-utility theorem identifies a rational ranking of lotteries with expected utility. The reward hypothesis identifies a goal for a policy with expected cumulative reward. The connection is that a policy induces a lottery over trajectories. If the utility of a trajectory is the sum of scalar rewards along it, the two statements are the same objective, $J(\pi)$.

The identification uses additivity and is not automatic. Part (f) is the counterexample: J_{average} is the expected utility of the trajectory averages, but it is not J_{instant}, the expected sum of the per-period logarithmic rewards. Turning J_{average} into a cumulative reward requires a terminal reward that depends on the whole history, and hence a state augmented by time and cumulative service.